# 崑山科技大學日間部電機工程系四技課程科目表

## Kun Shan University 4-Year College Curriculum (Day Division)

## 課程名稱(講授時數－實習時數－學分數)

## Course Name (Lecture Hours – Practice Hours – Credits)

113 學年度入學適用
Applicable to the freshmen enrolled in Fall, 2024
114.04.17 系課程委員會通過
114.04.22 院課程委員會通過
114.07.22 校課程委員會通過

| 第一學年 First Academic Year | | 第二學年 Second Academic Year | | 第三學年 Third Academic Year | | 第四學年 Fourth Academic Year | |
|---|---|---|---|---|---|---|---|
| 第一學期 First Semester | 第二學期 Second Semester | 第一學期 First Semester | 第二學期 Second Semester | 第一學期 First Semester | 第二學期 Second Semester | 第一學期 First Semester | 第二學期 Second Semester |
| **共同必修科目(合計 29 學分) University-wide Required Courses (29 Credits)** | | | | | | | |
| **校訂共同科目(合計 2 學分) General Courses (2 Credits)** | | | | | | | |
| | 生活實踐教育(一) Living Practice (I) (0-1-1) | 生活實踐教育(二) Living Practice (II) (0-1-1) | 服務學習與公民教育 Public Service and Civil Education (0-1-0) | | | | |
| **核心通識科目(合計 17 學分) Common Core Courses (17Credits)** | | | | | | | |
| 體育（一） Physical Education (I) (2-0-2) | 體育（二） Physical Education (I) (2-0-2) | | | 生涯發展與職業倫理 Career Development and Work Ethics (2-0-2) | 外語選讀 Foreign Languages ** (2-0-2) | | 英文能力檢定* English Proficiency Examination* (1-0-1) |
| 中文閱讀與表達(一) Chinese Reading and Expression(I) (2-0-2) | 中文閱讀與表達(二) Chinese Reading and Expression(II) (2-0-2) | | | | | | |
| 英文聽力與閱讀(一) English Listening and Reading (I) (2-0-2) | 英文聽力與閱讀(二) English Listening and Reading (II) (2-0-2) | | | | | | |
| **分類通識科目(合計 10 學分) Liberal Education Courses (10 Credits)** | | | | | | | |
| 分類通識 Liberal Education (2-0-2) | 分類通識 Liberal Education (2-0-2) | 分類通識 Liberal Education (2-0-2) | 分類通識 Liberal Education (2-0-2) | 分類通識 Liberal Education (2-0-2) | | | |
| 8-0-8 | 8-1-9 | 2-1-3 | 2-1-2 | 4-0-4 | 2-0-2 | | (1-0-1) |

註：Note:

*「英文能力檢定」依「崑山科技大學學生英文基本能力檢定實施辦法」辦理。

'English Proficiency Examination' is implemented according to 'Kun Shan University Guidelines for English Proficiency Requirement.'

**「外語選讀」包括「第二外語」及「職場英文」為選項式必修科目，課程擇一門修習即可。課程開設於三年級，然「第二外語」課程為二年級優先選課。
'Foreign Languages' comprises 'Foreign Languages (other than English)' and 'Workplace English', from which students choose one course. While the courses are offered in the third academic year, 'Foreign Languages (other than English)' is given preference to sophomores.

**院訂基礎必修科目（合計 15 學分）General Compulsory Courses of College (Total：15 Credits)**
**（機械、電機、電子、機器人）**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 微積分(一) (2-0-2) Calculus(I) | 微積分(二) Calculus(Ⅱ) (2-0-2) | 工程數學(一) Engineering mathematics (I) (3-0-3) | | | 學術倫理* Academic ethics (0-1-1) | | |
| 工程技術概論 Introduction to Engineering Technology (2-0-2) | 程式設計 Programming (1-2-3) | 科技英文 Technical English (2-0-2) | | | | | |
| 4-0-4 | 3-2-5 | 5-0-5 | 0-0-0 | 0-0-0 | 0-1-1 | 0-0-0 | 0-0-0 |

| 第一學年<br>First Academic Year | | 第二學年<br>Second Academic Year | | 第三學年<br>Third Academic Year | | 第四學年<br>Fourth Academic Year | |
|---|---|---|---|---|---|---|---|
| 第一學期<br>First Semester | 第二學期<br>Second Semester | 第一學期<br>First Semester | 第二學期<br>Second Semester | 第一學期<br>First Semester | 第二學期<br>Second Semester | 第一學期<br>First Semester | 第二學期<br>Second Semester |

**註*:「學術倫理」之實施與學分取得依課務組網頁之「學術倫理教育實施要點」辦理。The implementation of "Academic Ethics" and the acquisition of credits are handled in accordance with the "Implementation Points of Academic Ethics Education" on the website of the Academic Affairs Group.**

**院訂專業模組必修/必選科目（合計 21 學分）Professional Modules Compulsory/Required Courses of College (Total：21Credits)**

**機電整合專業模組課程（機械、電機、電子、機器人）**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 普通物理與實驗(一)<br>General Physics and Experiment (I)(2-1-3)(必修)<br>可程式控制器實習<br>PLC practice<br>(1-2-3) (必選) | 機器人工程師實務<br>Robot Engineer Practice (1-2-3)<br>(必選)<br>電路學<br>Electric Circuits<br>(3-0-3) (必修) | 電子學與實習<br>Electronics and practice<br>（1-2-3）(必修)<br>充電樁系統概論<br>Introduction to charging pile system（3-0-3）<br>(必選) | 車用晶片模組應用系統概論<br>Introduction to automotive chip module application system<br>（3-0-3）(必選) | | | | |
| 3-3-6 | 4-2-6 | 4-2-6 | 3-0-3 | 0-0-0 | 0-0-0 | 0-0-0 | 0-0-0 |

專業必修科目(合計 39 學分)Required subjects for majors (Total：39 Credits)

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 普通物理與實驗(二)<br>University Physics Experiment(Ⅱ)<br>(2-1-3) | 電機機械<br>Electric machinery fundamentals<br>(2-0-2)<br>電機機械實習<br>Electric Machinery Experiment<br>(0-2-2) | 進階電子學與實習<br>Advanced Electronics and practice<br>(1-2-3)<br>半導體產業技術<br>Semiconductor Industry Technology<br>(3-0-3)<br>工程數學(二)<br>Engineering mathematics<br>(Ⅱ)<br>(3-0-3) | 實務專題(一)<br>Practice Project<br>(Ⅰ)<br>(0-3-3)<br>電力電子學<br>Power Electronics<br>(2-0-2)<br>電力電子學實習<br>Experiments of Power Electronics<br>(0-2-2)<br>基礎量測及實習<br>The Fundamental of Measurement and Experiment<br>(1-2-3) | 實務專題(二)<br>Practice Project<br>(II)<br>(0-3-3)<br>控制系統<br>control systems<br>(2-0-2)<br>控制系統實習<br>control systems Experiment<br>(0-2-2) | | 就業與創業技能輔導<br>Employment and entrepreneurial skills counseling<br>(3-0-3)<br>專業證照*<br>Professional Licenses<br>(1-0-1) | 暑期校外實習**<br>Summer Internship<br>(0-＊-2) |
| 0-0-0 | 2-1-3 | 2-2-4 | 7-2-9 | 3-7-10 | 2-5-7 | 0-0-0 | 4-0-4 | 0-*-2 |

註*：專業證照課程依電機系之「專業證照課程實施辦法」辦理。Professional certificate courses are handled in accordance with the "Implementation Measures for Professional Certificate Courses" of the Department of Electrical Engineering.

註**：暑期校外實習課程依電機系之「校外實習課程實施辦法」辦理。Summer off-campus internship courses are handled in accordance with the "Implementation Measures for Off-campus Internship Courses" of the Department of Electrical Engineering.

專業選修科目，依每學期實際情況開課（選修學分 26 學分, 本系至少 17 學分）

Major elective subjects are taught according to the actual conditions of each semester (29 credits for electives, at least 20 credits for this department)

能源科技學程 Energy Technology Program：

| 第一學年<br>First Academic Year | | 第二學年<br>Second Academic Year | | 第三學年<br>Third Academic Year | | 第四學年<br>Fourth Academic Year | |
|---|---|---|---|---|---|---|---|
| 第一學期<br>First Semester | 第二學期<br>Second Semester | 第一學期<br>First Semester | 第二學期<br>Second Semester | 第一學期<br>First Semester | 第二學期<br>Second Semester | 第一學期<br>First Semester | 第二學期<br>Second Semester |
| | 可程式控制器應用及實習<br>Practice of Programmable Logic Controller and Internship<br>(1-2-3) | 工業配線及實習<br>Industrial Wiring and Experiment<br>(1-2-3) | 電機機械專論<br>Advanced of Electric Machinery<br>(3-0-3)<br>綠色能源概論<br>Introduction to Green Energy<br>(3-0-3) | 電力系統<br>Power System<br>(3-0-3) | 電力系統分析<br>Power System Analysis(3-0-3)<br>發變電工程<br>Engineering of Power Generation and Transformation<br>(2-0-2)<br>配電工程<br>Power Distribution System<br>(2-0-2)<br>電動機控制與實務<br>Motor Control and Practice<br>(3-0-3)<br>充電樁規劃與設計 Charging pile planning and design<br>(3-0-3) | 電力品質分析<br>Electric Power Quality Analysis<br>(3-0-3)<br>電力系統模擬<br>Power System Simulation<br>(3-0-3)<br>風力發電技術<br>Wind Power Technology<br>(3-0-3)<br>電力控制與能源管理<br>Power Control and Energy Management<br>(3-0-3)<br>充電樁製造生產實務 Charging pile manufacturing and production practices<br>(2-2-4) | 能源應用<br>Energy applications<br>(3-0-3)<br>電力電子應用<br>Applications of Power Electronics<br>(3-0-3)<br>太陽光電能發電系統<br>Photovoltaic Energy Generation System<br>(3-0-3)<br>充電樁測試檢驗實務<br>Charging pile testing and inspection practices<br>(2-2-4) |
| 電機系統控制學程 Electrical Machine System Control Course： | | | | | | | |
| | 可程式控制器應用及實習<br>Practice of Programmable Logic Controller and Internship<br>(1-2-3) | 可程式控制器進階控制實習<br>Advanced Control and Experiment of Programmable Logic Controller<br>(0-3-3) | 可程式 LabView 圖形監控<br>LabView Visual Monitoring of Programming Logic Con<br>(3-0-3) | 數位影像處理<br>Digital Image Processing<br>(3-0-3) | 影像視覺檢測<br>Image visual inspection<br>(3-0-3) | 可程式伺服控制及實習<br>servo-control and practice<br>(1-2-3) | 感測元件應用與實習<br>Sensing component application and internship<br>(1-2-3) |
| | 機電整合基礎實習<br>Fundamentals of Mechatronics with Practice<br>(0-3-3) | | | 機電整合應用及實習<br>Electromechanical application and practice<br>(1-2-3) | 機電整合實務<br>Mechatronics Practice<br>(3-0-3) | 真空半導體設備<br>人機介面及實習<br>Practice on human interface of vacuum equipment<br>(1-2-3) | PC Based 自動化控制實習<br>PC Based Automatic Control and Practice<br>(1-2-3) |
| | | | | VB 監控實習<br>VB monitoring practice (0-3-3) | 單晶片實務應用<br>Practical application of single-chip microprocessor<br>(3-0-3) | 微電腦介面應用<br>Microcomputer Interface Applications<br>(3-0-3) | |
| 半導體工程學程 Semiconductor Engineering Program： | | | | | | | |

| 第一學年<br>First Academic Year | | 第二學年<br>Second Academic Year | | 第三學年<br>Third Academic Year | | 第四學年<br>Fourth Academic Year | |
|---|---|---|---|---|---|---|---|
| 第一學期<br>First Semester | 第二學期<br>Second Semester | 第一學期<br>First Semester | 第二學期<br>Second Semester | 第一學期<br>First Semester | 第二學期<br>Second Semester | 第一學期<br>First Semester | 第二學期<br>Second Semester |
| 材料科學導論<br>Fundamentals of Materials Science and Engineering<br>(3-0-3) | 奈米科技導論<br>Introduction to Nanotechnology<br>(2-0-2) | AI 半導體製造概論<br>Introduction to AI Semiconductor Manufacturing<br>(3-0-3) | 半導體綠能科技應用<br>Practice on human interface of vacuum equipmen<br>(3-0-3) | 半導體元件<br>Semiconductor devices<br>(3-0-3) | 真空技術<br>Vacuum technology<br>(3-0-3) | 光電元件<br>Opto-Electronic Devices<br>(3-0-3) | AI 智慧照明<br>AI Smart Lighting<br>(3-0-3) |
| | | | 光電工程導論<br>Introduction to Optoelectronic Engineering<br>(3-0-3) | 半導體製程<br>Semiconductor Processing<br>(3-0-3) | 薄膜製程<br>Thin Film Process<br>(3-0-3) | AI 感測器元件<br>AI Sensor Element<br>(3-0-3) | 薄膜技術實務<br>Thin Film Technology Practice<br>(3-0-3) |
| | | | | 太陽能電池<br>Solar Cell<br>(3-0-3) | 高亮度 LED 技術<br>High brightness LED technology<br>(3-0-3) | Micro LED 照明技術<br>Micro LED lighting technology<br>(3-0-3) | 真空半導體設備人機介面及實習<br>Practice on human interface of vacuum equipment<br>(1-2-3) |
| 其它選修 Other electives： | | | | | | | |
| 電機概論<br>Introduction to electrical engineering<br>(2-0-2) | 邏輯設計<br>Logic Design<br>(2-0-2) | 邏輯設計實習<br>Logic Design Experiment<br>(0-2-2) | 電磁學概論<br>General Electromagnetics<br>(2-0-2)<br>微處理機基礎<br>Basis of Microprocessor<br>（1-2-3） | Matlab 工程軟體應用<br>Application of Matlab Engineering Software<br>(3-0-3) | 數位電子乙級技術士學科<br>Digital electronic grade B technique master theor<br>(3-0-3) | 數位電子乙級技術士術科<br>Digital Electronics grade B technique master technique<br>(1-2-3) | 專利實務<br>Patent practice<br>(3-0-3) |
| 計算機概論<br>Introduction to Computer<br>(2-0-2) | C++程式語言<br>C++ Programming Language<br>(3-0-3) | C#程式語言<br>C# Programming Language (3-0-3) | 網路分析<br>Network Analysis<br>(3-0-3) | LabView 工程軟體應用<br>Computer control software applications<br>(3-0-3) | 電力電子乙級技術士學科<br>**Academic Study for the Test of** Level B Technician for Power Electronics<br>(3-0-3) | 電力電子乙級技術士術科<br>**Skill subjects of** Level B Technician for Power Electronics<br>(1-2-3) | 創意與專利<br>Creation and Patent<br>(3-0-3) |
| | | | | 資料結構<br>Data Structures<br>(3-0-3) | 機電整合乙級技術士學科<br>Mechatronics grade B technique master theoretical course<br>(3-0-3) | 機電整合乙級技術士術科<br>Mechatronics grade B technique master technique<br>(1-2-3) | 職場實習<br>Workplace internships<br>(0-＊-7) |

| 第一學年<br>First Academic Year | | 第二學年<br>Second Academic Year | | 第三學年<br>Third Academic Year | | 第四學年<br>Fourth Academic Year | |
|---|---|---|---|---|---|---|---|
| 第一學期<br>First Semester | 第二學期<br>Second Semester | 第一學期<br>First Semester | 第二學期<br>Second Semester | 第一學期<br>First Semester | 第二學期<br>Second Semester | 第一學期<br>First Semester | 第二學期<br>Second Semester |
| | | | | | 校外實習<br>Off-campus internship<br>(0-＊-2) | 學期校外實習(一)<br>Semester Intership (I)<br>(0-＊-9) | 學期校外實習(二)<br>Semester Intership (Ⅱ)<br>(0-＊-9) |
| | | | | | | 實務專題(三)<br>Practice Project (Ⅲ)<br>(0-1-1) | 實務專題(四)<br>Practice Project (Ⅳ)<br>(0-1-1) |

註：

校外實習課程依電機系之「校外實習課程實施辦法」辦理

Off-campus internship courses are handled in accordance with the "Implementation Measures for Off-campus Internship Courses" of the Department of Electrical Engineering

## 最低畢業總學分為 130 學分 The minimum total credits for graduation is 130 credits

**註：**

**1.本系最低畢業學分為 130 學分：必修 92 學分，必選 12 學分，選修 26 學分。The minimum graduation credits of this department are 130 credits: 92 credits of compulsory courses, 12 credits of required elective, and 26 credits of electives.**

**2.本系學生需至少修讀完成一跨院系學分學程，取得跨院系學分學程證書使得畢業。 Students in this department need to complete at least one cross-department credit course and obtain a cross-department credit course certificate to graduate.**

**3.本系學生需至少修讀完成一數位科技微學程，取得數位科技微學程證書使得畢業。 Students in this department need to complete at least one digital technology micro-course and obtain a digital technology micro-course certificate to graduate.**

**4.選修跨院系學程之外系專業課程，得等同本系之專業選修學分。Elective professional courses outside the department's cross-department program will be equivalent to the professional elective credits of the department.**

**5.暑期校外實習之實施為 4 週 160 小時 2 學分於暑假期間實施，若超過 160 小時，第 161-320 小時可抵校外實習選修 2 學分，未滿 160 小時不可抵免；實習期滿成績及格者方給予承認課程並抵免學分。學期校外實習之實施為開設 9 學分，至少為期 4.5 個月之校外實習課程，修讀實習課程期間，除依各校訂定定期返校之座談會或研習活動等外，學生應全職於實習機構實習。若於四年級選修學期校外實習課程,則可抵免暑期校外實習及專業選修 7 學分。The implementation of summer off-campus internship is 160 hours and 2 credits in 4 weeks during the summer vacation. If it exceeds 160 hours, the 161-320 hours can be used as 2 credits of off-campus internship. Less than 160 hours cannot be deducted. Only those who have passed the internship period can Grant recognized courses and credit. The implementation of off-campus internship during the semester is an off-campus internship course with 9 credits and a minimum duration of 4.5 months. During the internship course, in addition to regular symposiums or study activities determined by each school, students should practice full-time in the internship institution. . If you take a semester off-campus internship course as an elective in your fourth year, you can deduct 7 credits of summer off-campus internship and professional elective courses.**